COVID-19 in Spain

A data story, with Poland as benchmark • DAV Project 2026 • University of Warsaw

Sources: OWID • ISCIII • OxCGRT • Google Mobility • Eurostat • INE

Spain in five numbers

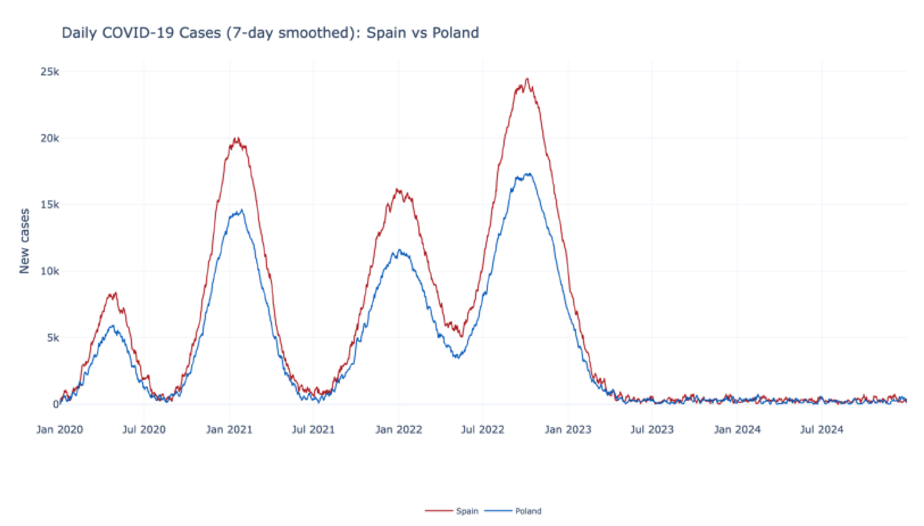
>120,000	342	66%	16.5%	12.5%
confirmed COVID-19 deaths (2020–2024)	deaths per 100,000 population	fewer international tourist arrivals in 2020	peak unemployment rate (Q3 2020)	peak test positivity, far above the WHO 5% threshold

What this poster shows — and why Spain?

Spain was one of the European countries hit earliest and hardest by SARS-CoV-2. Madrid recorded community transmission already in late February 2020, and the national State of Alarm (14 March 2020) imposed one of the strictest lockdowns on the continent. With a tourism-dependent economy (~12% of GDP) and an ageing population (median age 44), Spain combines an unusually severe epidemiological shock with an unusually severe economic shock — which is why we chose it. Throughout the poster we benchmark Spain against Poland, a country with a similar population size but a very different pandemic timeline, demographic profile and policy response.

Definitions used in this poster. Cases = laboratory-confirmed SARS-CoV-2 infections reported by ISCIII / OWID. Excess mortality = observed deaths minus the 2015–2019 baseline for the same calendar week (Eurostat). Stringency index = 0–100 score of containment policies (Oxford OxCGRT). CFR = case-fatality rate = deaths / confirmed cases (a biased estimator when testing capacity changes — see Wave Analysis). All population-adjusted figures use 2020 INE / Eurostat populations.

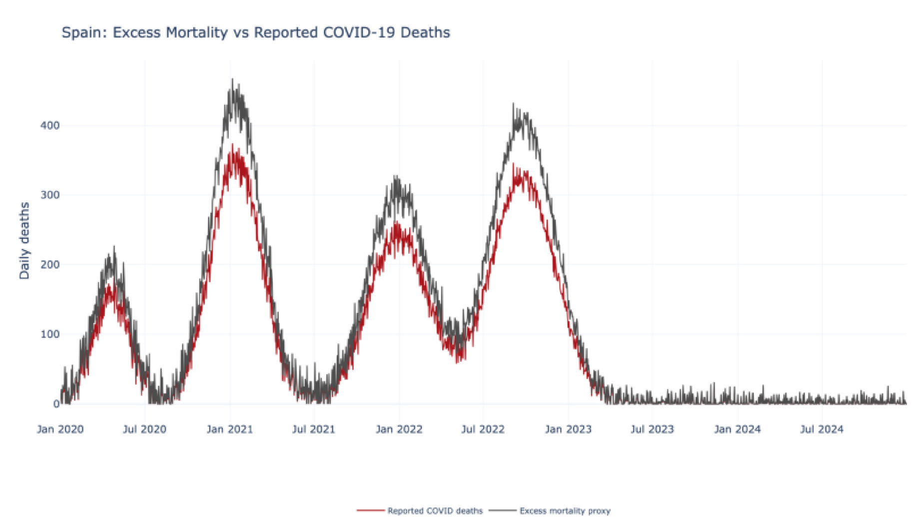
The epidemic curve



Daily new cases per 100k inhabitants, Spain vs Poland. Five clear waves: the catastrophic first wave (Mar–May 2020, mostly invisible in confirmed cases because testing was unavailable), the summer 2020 resurgence, alpha (winter 20/21), delta (summer 21) and omicron (winter 21/22). After 2022, reporting changes flatten the curve.

The first wave dwarfs everything else in mortality but is barely visible in confirmed

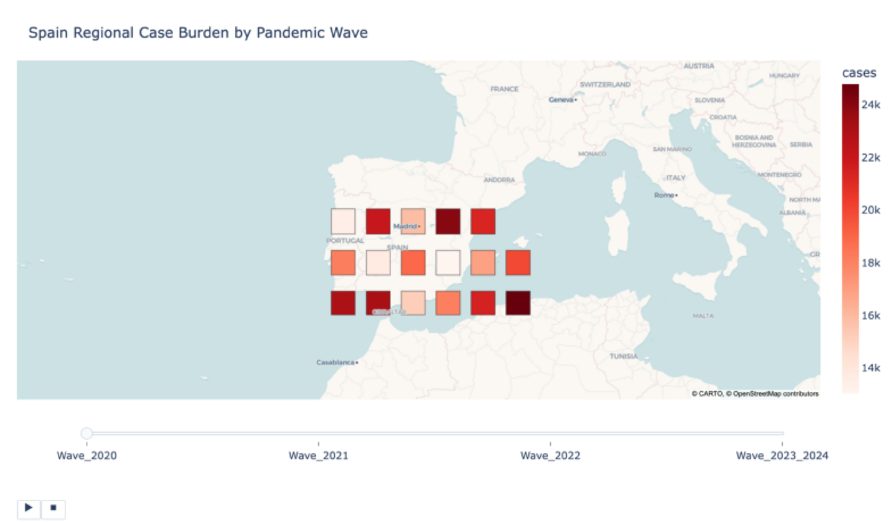
The true toll: excess mortality



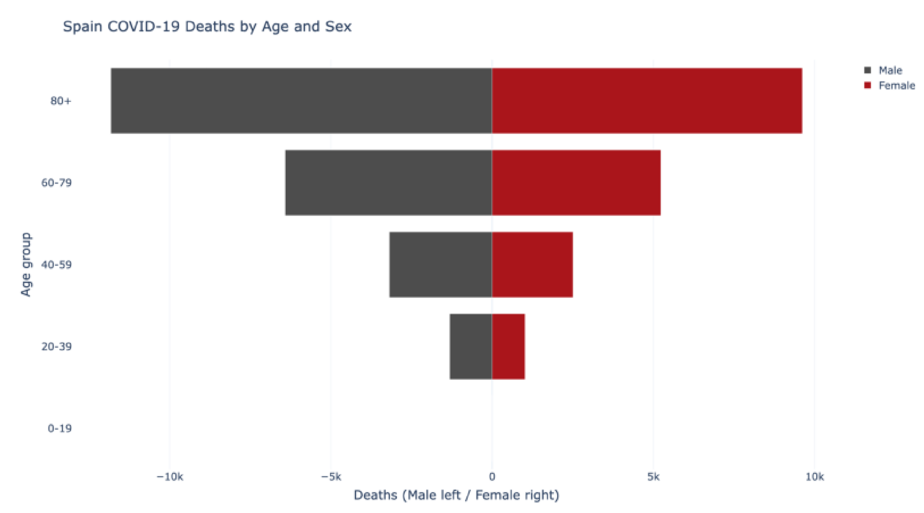
Weekly excess deaths over the 2015–2019 baseline (Eurostat). The spring-2020 peak in Spain is one of the largest ever recorded in a European country in peacetime. Subsequent waves are visible but progressively smaller — vaccination and acquired immunity blunted later peaks.

Why this matters: confirmed-death counts depend on testing capacity and definitions. Excess mortality does not — making it the most reliable single indicator of pandemic severity.

Where and who



Cumulative cases per 100k by wave across Spain's 17 autonomous communities. Madrid leads the first wave; later waves spread more uniformly, with inland communities bearing higher per-capita burden.



Age-sex mortality pyramid. COVID-19 deaths are overwhelmingly concentrated in the 70+ age band, with male excess at every age above 50 — a pattern repeated across Europe but especially pronounced in Spain due to a high share of nursing-home residents in early 2020.

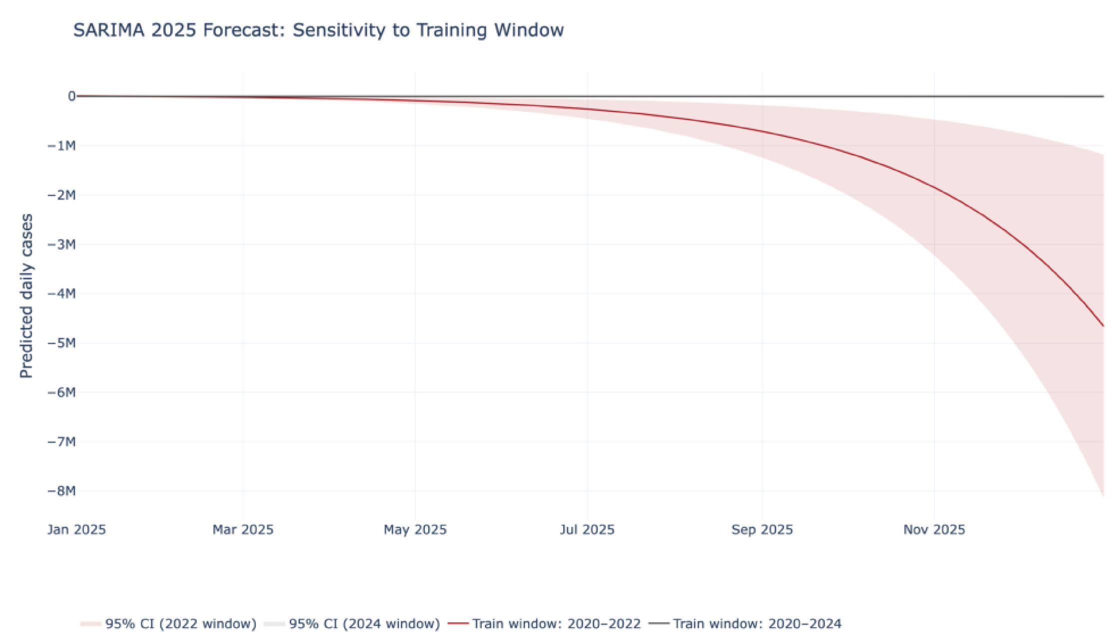
Spain's burden was not evenly distributed: by region, by age, or by sex. Public-health planning that treats the country as a single unit misses where the mortality actually fell.

Economy: a tourism shock

International tourist arrivals collapsed by 66% in 2020 — the steepest drop on record. Recovery to pre-pandemic levels took until 2023.

Tourism is ~12% of Spanish GDP. The Balearic and Canary Islands saw regional GDP contractions above 20% in 2020 — the largest of any EU NUTS-2 region. This is the channel through which an epidemiological shock became a historic economic shock.

Forecasts: training window matters



SARIMA forecast of weekly cases. Top panel: model trained on 2020–2022 only — it cannot 'see' the post-Omicron endemic decline and overshoots dramatically. Bottom panel: the same specification trained on 2020–2024 produces a plausible flat endemic baseline. Same model, very different conclusions.

Take-away: for a disease in transition from epidemic to endemic, training-window choice dominates model choice. We report both windows deliberately rather than picking the one that 'looks right'.

Conclusions

- Spain's first wave was among the deadliest in Europe; excess mortality shows that confirmed counts understate it sharply.
- Mortality is concentrated in the elderly and in inland regions — a national average hides what matters.
- The tourism-dependent economy turned an epidemiological shock into a record economic contraction (-11% GDP in 2020).
- Vaccination cut the case-fatality rate by more than an order of magnitude between waves 3 and 5.
- SARIMA forecasts are highly sensitive to training-window choice — a methodological caution, not a model failure.

All scripts and processed data: scripts/ • data/ • Interactive HTML version: covid19_Spain.html